Does a pond-scale thermal index travel?

Testing a spring peeper calling-onset model against continental monitoring data

UNIVERSITY OF GUELPH

Farah Sadoon, Karl Cottenie

BACKGROUND

Phenological events are shifting as the climate warms, and amphibians have significantly advanced their breeding[1].

Anuran calling behaviours are sensitive to both within-season climate anomalies such as false springs[2], and long-term climate warming[3]. Since their calls are distinct and easily monitored, anuran calling events can be used to track the effects of climate change and model calling behaviours.

However, we need to identify which specific cues species or populations are responding to for modelling their calling behaviour effectively.

Spring peepers (*Pseudacris crucifer*) are a useful study system because they occupy most of eastern North America. Their range spans ~26 degrees of latitude from the Gulf states to southern Labrador and west to Manitoba[4]. Their expansive occupation provides an opportunity to assess the effectiveness of cues as predictors under different thermal regimes.

Lovett[5] conducted a study of spring peeper date of first calling (DFC) at a single pond in New York and identified that the best indicator of calling onset was a thermal sum accumulated from 1 February with a base of 3°C, reaching a threshold of 44.3 growing degree-days (TS3); however, we do not know if this threshold is a species-wide physiological constant, or a population-specific one.

N. Cairns, Flickr

Motivating Questions:

(1) Is the TS3 cue informative for predicting DFC at sites across the spring peeper range, beyond Lovett's pond?

(2) Does TS3 add information for predicting calling beyond the site's geography?

METHODS

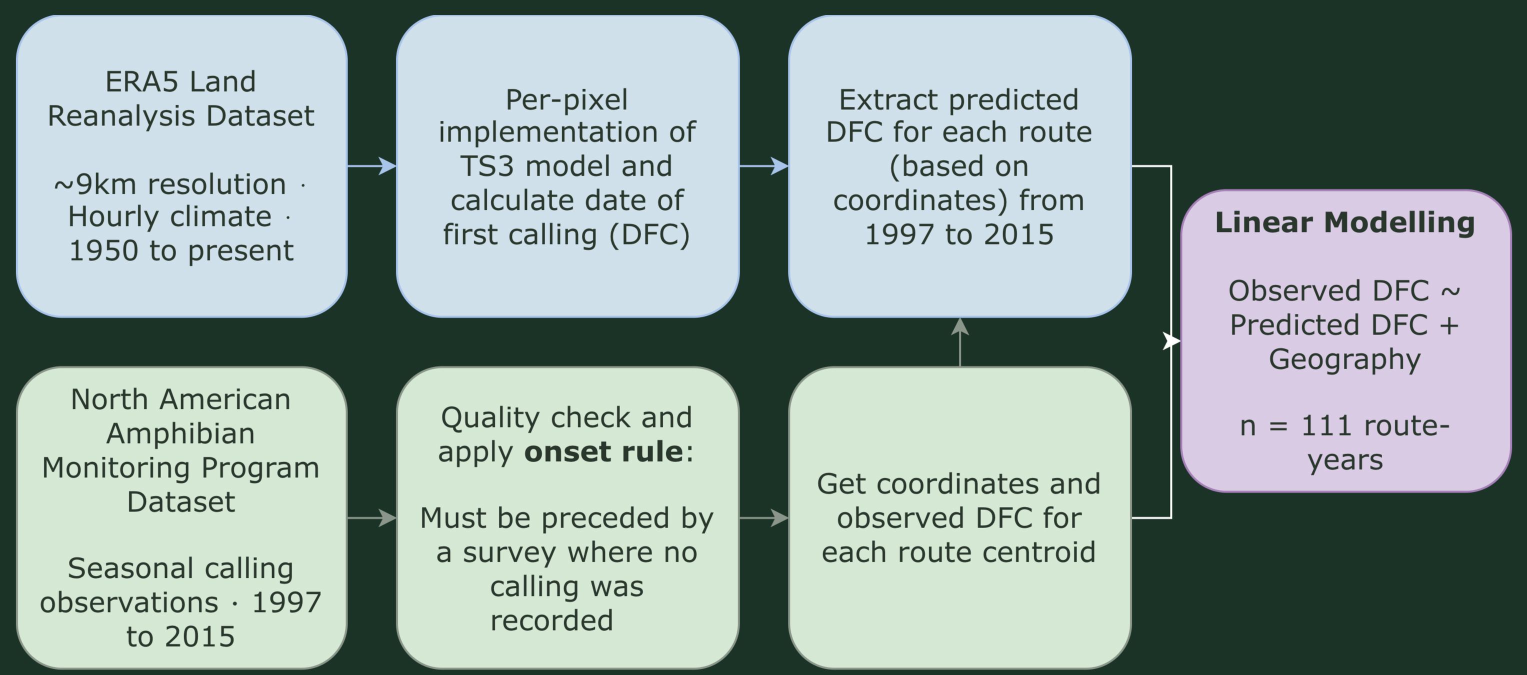

Figure 1. Modelling workflow. Climate data processed in Google Earth Engine (top, blue) and NAAMP survey data quality-controlled processed in R (bottom, green) were combined to model observed DFC as a function of TS3 and geography (middle, purple).

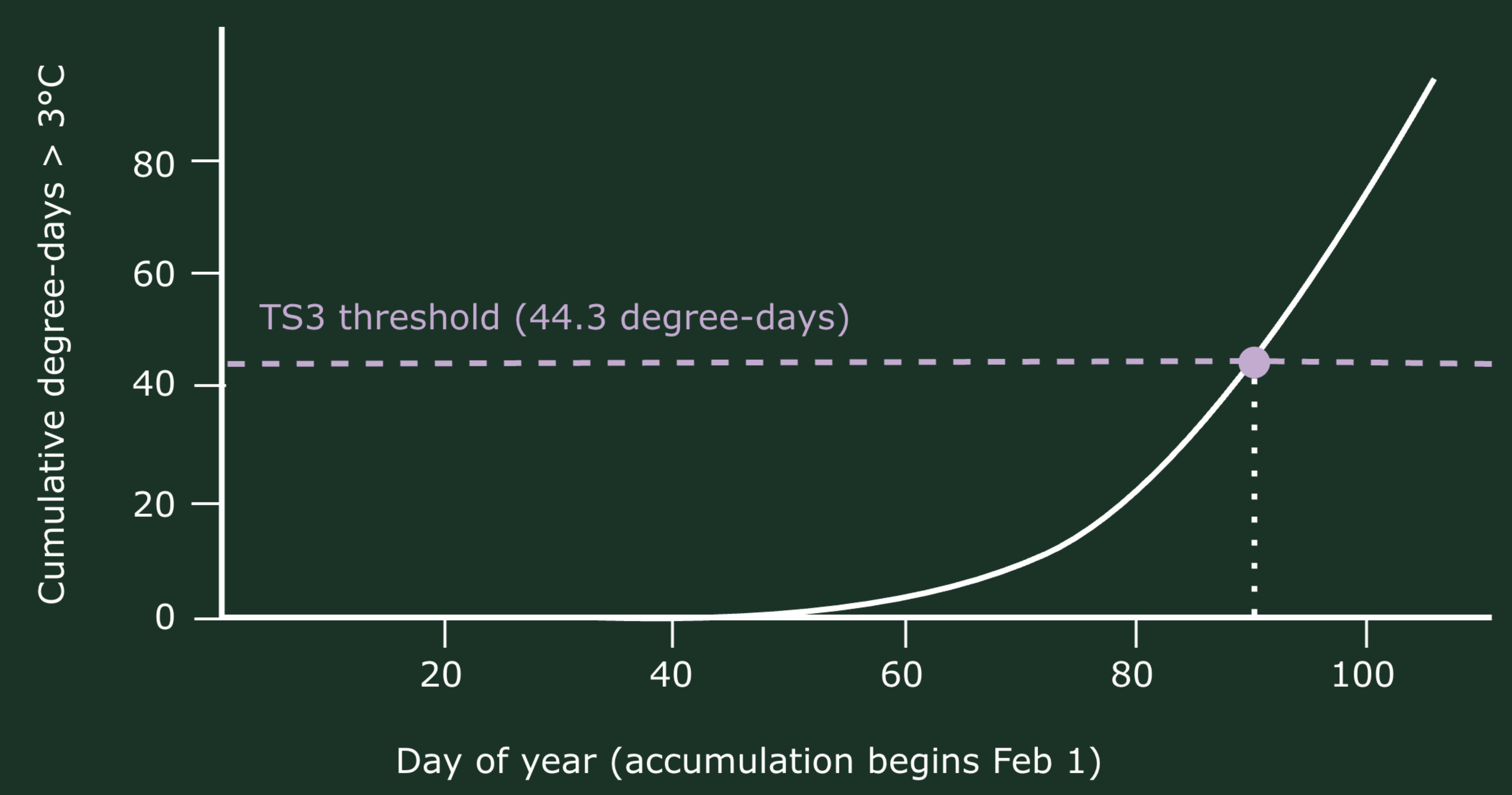

Figure 2. Graphical representation of Lovett's TS3 model. Dashed line represents the TS3 threshold of 44.3 growing degree days at a base of 3°C, and dotted line represents the day of year which the threshold was crossed. The window for this model starts at February 1, or Julian day 32. Here, predicted DFC is 90.

RESULTS

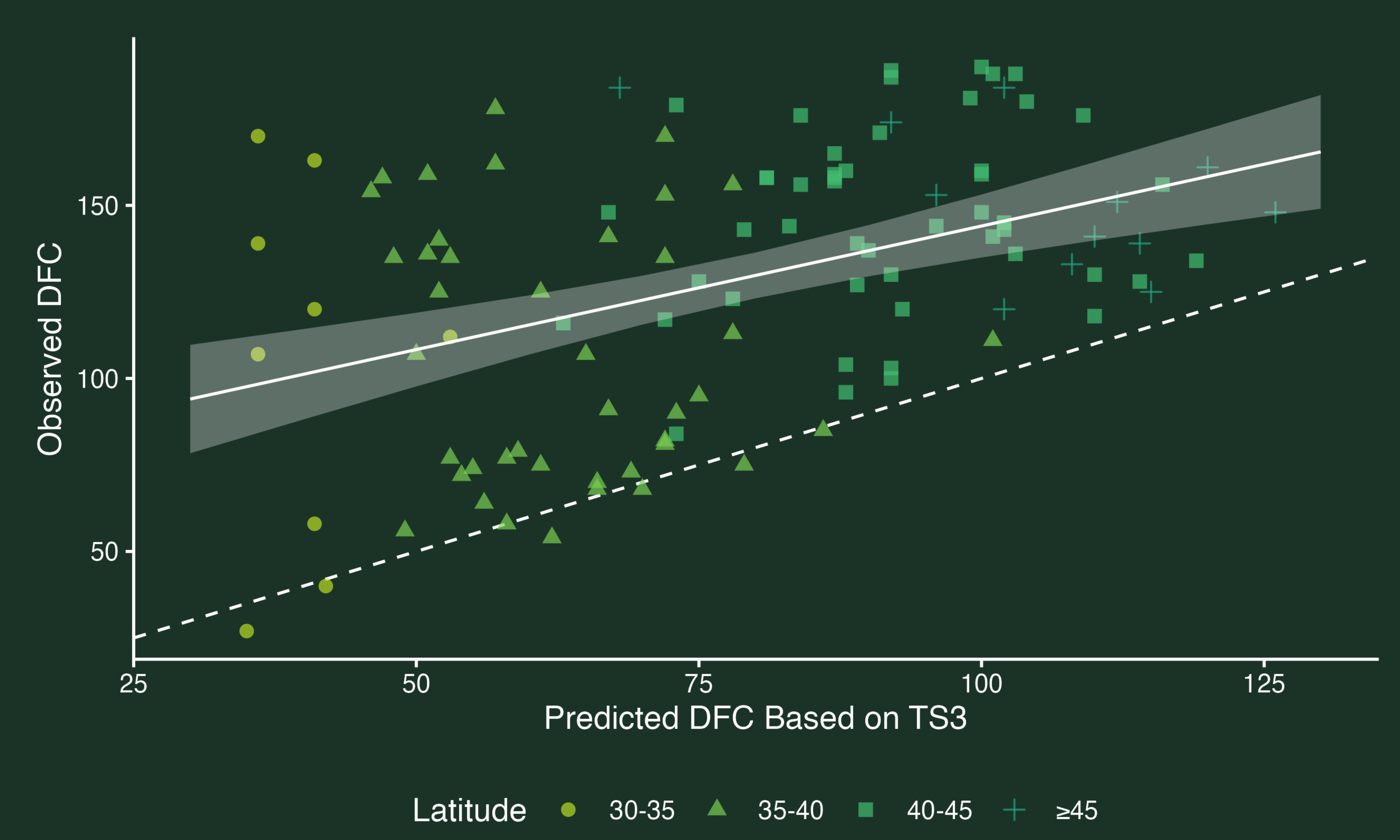

Figure 3. Observed calling onset against TS3-predicted onset at 90 NAAMP routes (111 route-years) by latitude band. TS3 significantly predicted observed onset (slope = 0.71, $p < 0.001$), but observations occurred systematically later than predicted (intercept = 72.6 days). Dashed line indicates 1:1 agreement.

Table 1. Comparison of candidate models predicting observed calling onset. Geography alone outperformed TS3 alone (ΔAIC = 27.2), and adding TS3 to geography produced a small improvement ($F_{1,107}$ = 5.85, p = 0.017). ΔAIC is relative to the best-supported model (*).

Model	*k*	*R²*	*RSE*	*ΔAIC*
TS3	2	0.177	35.1	31.1
TS3 + year	3	0.234	34.0	25.2
Latitude + Longitude	3	0.368	30.9	3.91
TS3 + Latitude + Longitude*	4	0.400	30.2	0
TS3 x Latitude + Longitude	5	0.400	30.4	1.97

Table 2. Coefficients from the best-supported model (observed DFC ~ TS3 + latitude + longitude; R^2 = 0.400, n = 111 route-years). Latitude and longitude are centred on their means. Holding TS3 constant, onset occurred approximately 9.6 days later per degree of latitude north. Holding location constant, each additional day of delay in reaching the TS3 threshold was associated with onset approximately 0.6 days earlier.

Term	*Estimate*	*SE*	*p*
Intercept	176	19.8	< 0.001
Predicted DFC (TS3)	-0.603	0.249	0.017
Latitude (days per °N)	9.64	1.56	< 0.001
Longitude (days per °E)	1.26	0.374	0.001

DISCUSSION

Here, we show that TS3 transfers beyond Lovett's pond[5] to sites across the spring peeper range (Figure 3), explaining 17% of variance in observed DFC (Table 1).

This estimate indicates that factors beyond thermal accumulation influence variation in calling onset. Geographic position alone explained more variance than TS3 (37%), and the best-supported model combined both (40%, Table 1).

The best model indicates that when TS3 and longitude are held constant, a one-degree increase in latitude is associated with an approximately 9-day delay in calling (Table 2). Because thermal accumulation is accounted for, this suggests that factors covarying with latitude, but not with accumulated warmth, contribute to onset timing.

- **Photoperiod**: day length varies with latitude but is unaffected by warming. Recent work shows that day of year predicts calling independently of temperature rainfall in Australian frogs examined at continental scale[6].
- **Snowmelt**: winter conditions drive organism responses differently than spring warmth[7] and could contribute to the presence of ephemeral ponds.

After accounting for geography, TS3 retained a negative slope: when thermal threshold was reached one day later, onset occurs 0.6 days earlier (Table 2). This is inconsistent with the direction implied by a thermal-sum mechanism.

- **Strict tracking may be costly**: delaying calling may be costly due to competition, and calling to early may expose peepers to frost risk or a false-spring effect observed in other species[2,8].

ACKNOWLEDGEMENTS & REFERENCES

Thank you to Dr. Karl Cottenie for his enthusiasm and support throughout this project, which sparked my own excitement and played an important role in shaping its direction. I am also grateful to Dr. Amanda Cicchino, whose insight and inspiration first planted the idea of modelling frog phenology and whose guidance and encouragement were invaluable.

1. Parmesan. (2007). *Global Change Biology*
2. Ethier et al. (2026). *Integrative Zoology*
3. Gibbs & Breisch. (2001). *Conservation Biology*
4. IUCN SSC Amphibian Specialist Group. (2022). *Pseudacris crucifer*
5. Lovett. (2013). *Northeastern Naturalist*
6. Thomspon et al. (2022). *Diversity and Distributions*
7. Williams et al. (2015). *Biological Reviews*
8. Bison et al. (2021). *Front. Ecol. Evol.*